TakhleeqX

Project Team

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Chapter 1

Introduction

The purpose of this project proposal document is to present the concept, objectives, and implementation plan for TakhleeqX, an AI-powered autonomous marketing agent system. It defines the scope, tools, workflow, and expected outcomes of the project, while also highlighting its innovation compared to existing marketing automation tools. This document serves as a foundation for development, evaluation, and defense of the project.

Background of the System:

Digital marketing is critical for businesses today, but managing campaigns across multiple platforms is costly, time-consuming, and requires specialized expertise. Existing tools like Jasper AI and ActiveCampaign provide partial solutions — Jasper focuses on content creation, while ActiveCampaign automates email workflows. However, both still depend heavily on human intervention.

TakhleeqX addresses this gap by providing a closed-loop autonomous marketing system. Using LangChain, LLMs, and integrated APIs, it automates the entire pipeline — from trend analysis, content creation, and ad strategy planning to campaign deployment, customer interaction, and performance analytics. Unlike traditional tools, it is designed to run with minimal human input, continuously optimizing campaigns through feedback loops. [1].

1.1 Existing Solutions

In the realm of digital marketing automation and AI-driven content generation, several platforms have been developed to streamline marketing efforts. Two prominent existing solutions are ActiveCampaign and Jasper AI.

ActiveCampaign is a comprehensive marketing automation and customer relationship management (CRM) platform. It provides tools for email marketing, sales automation, and lead scoring. While highly effective for managing workflows and customer journeys, it primarily focuses on rule-based automation and lacks native, autonomous generative AI capabilities for creating end-to-end campaigns from scratch. Users must still manually conceptualize campaigns and write the content or integrate third-party tools to do so.

Jasper AI (formerly Jarvis) is an advanced AI writing assistant designed to generate high-quality marketing copy, blog posts, and social media content. It leverages large language models to produce text based on user prompts. However, Jasper AI operates primarily as an isolated content generation tool. It lacks the architectural framework to autonomously scout real-time market trends, visually design corresponding graphics, and orchestrate the publishing pipeline without continuous human intervention.

These existing solutions address specific silos of the marketing process but fail to provide a cohesive, autonomous system that seamlessly connects trend analysis, strategic planning, multimodal content generation, and publishing. TakhleeqX addresses these gaps by utilizing a stateful multi-agent architecture that fully automates the campaign lifecycle from trend discovery to content publishing.

Table 1.1: Comparison of Existing Solutions

System Name	**System Overview**	**System Limitations**
ActiveCampaign	A comprehensive CRM and marketing automation tool for managing customer journeys and email workflows.	Relies on rule-based automation; lacks autonomous generative AI for end-to-end content creation and strategy.
Jasper AI	An AI-powered writing assistant that generates high-quality marketing copy and text content.	Acts as an isolated tool; lacks real-time trend scouting, graphic design generation, and autonomous orchestration.
TakhleeqX	A multi-agent AI marketing platform that automates trend scouting, strategy planning, content creation, and publishing.	Currently operates in a simulated environment; relies on external APIs (OpenAI/DALL-E) which can incur costs.

1.2 Problem Statement

Marketing has become a necessity for every brand in today's digital world, as businesses rely heavily on online presence to attract and retain customers. However, most marketing agencies are either too expensive or outdated in their strategies, making them inaccessible for small businesses and startups. Traditional marketing requires constant human effort for trend analysis, content creation, ad planning, posting, and performance tracking, which consumes time and resources. Even when businesses use automation tools like ActiveCampaign or content generators like Jasper, they still face the challenge of manual intervention for strategy, deployment, and optimization. This results in inefficiency, increased costs, and limited scalability.

Furthermore, businesses often lack data-driven insights to understand market trends and customer preferences, leading to generic campaigns that fail to connect with the target audience. Human-driven marketing also struggles with real-time adaptation — if customer behavior changes, campaign strategies cannot be adjusted quickly without additional cost and effort. These limitations create a gap in the market for an intelligent, autonomous system that can handle the end-to-end marketing cycle.

Our system, TakhleeqX, solves this problem by combining AI agents with automation, enabling businesses to run campaigns with minimal human input. It not only creates tailored content and ad strategies but also posts them automatically, engages with customers, and measures performance to continuously optimize results. By reducing reliance on manual labor and expensive agencies, TakhleeqX makes advanced digital marketing accessible, adaptive, and cost-effective for businesses of all sizes.

1.3 Scope

The scope of this project is to design and develop an AI-powered autonomous marketing system, named TakhleeqX, that automates the complete digital marketing cycle for businesses. The system will cover the major stages of marketing, including trend analysis, content creation, ad strategy planning, automated posting, customer engagement, and performance analytics. Within its boundaries, the system will rely on pre-trained language and vision models to generate marketing assets, APIs for social media and ad deployment, and data-driven insights for optimization. The project will specifically focus on businesses such as restaurants, startups, and SMEs that lack the resources to hire expensive marketing agencies or specialized teams.

The system will begin by analyzing market data, customer reviews, and social media trends to detect emerging opportunities. Based on these insights, it will generate creative assets such as ad copy, posters, and promotional content aligned with the target audience's preferences. The ad strategy module will determine the most suitable platforms, audience segments, budgets, and campaign schedules. Once the campaign plan is finalized, the automated posting agent will publish content across platforms like Facebook, Instagram, and WhatsApp while also interacting with customers through chatbots.

The performance analytics module will track key metrics such as clicks, conversions, ROI, and customer sentiment to measure success and provide feedback into the system. This ensures the software operates as a closed-loop cycle, where insights from one campaign are used to improve the next. The project will not attempt to build new foundation models from scratch but will instead integrate existing pre-trained models, ensuring feasibility and faster deployment. It will also remain limited to digital platforms, meaning offline marketing such as TV or billboard advertising is outside its scope.

The boundaries of this project are thus defined by autonomous campaign generation and optimization using AI agents, APIs, and analytics. Its main functionalities include reducing manual work, lowering costs, providing personalized marketing strategies, and adapting campaigns in real time. In summary, the project will deliver a scalable, adaptive, and cost-effective autonomous marketing solution that bridges the gap between expensive agencies and limited traditional automation tools.

1.4 Modules

1.4.1 Business Onboarding Module

This module enables businesses to register on the platform and provide key details about their services, menu, and goals.

1. Collects business details such as name, menu, services, and offers.
2. Allows users to define marketing goals (brand awareness, sales increase, customer engagement).
3. Stores business preferences for personalization of campaigns.

1.4.2 Trend Analysis Agent Module

This module analyzes real-time market data, social media activity, and competitor performance to identify trending topics and customer preferences.

1. Gathers data from social media, reviews, and competitor ads.
2. Identifies trending topics, high-demand products, and customer preferences.
3. Generates market insights and passes them to the content creation module.

1.4.3 Content Creation Agent Module

This module generates ad creatives such as captions, posters, reels, and other promotional material using AI models.

1. Produces ad creatives including captions, posters, reels, and menu cards.
2. Customizes content based on trend insights and business inputs
3. Supports text and image generation using AI models.

1.4.4 Ad Strategy Agent Module

This module designs optimized ad campaigns by selecting audiences, platforms, budgets, and strategies for maximum impact.

1. Selects the best audience segments (e.g., students, families, office workers).
2. Chooses platforms (Facebook, Instagram, WhatsApp) for campaign deployment.
3. Sets campaign parameters such as budget, duration, and bidding strategy.

1.4.5 Automated Posting Agent Module

This module automates the scheduling and posting of campaigns across platforms, ensuring timely promotions and customer engagement.

1. Schedules and posts campaigns across selected platforms.
2. Automates daily offers and time-specific promotions.
3. Integrates chatbot functionality for responding to customer queries via DMs or WhatsApp.

1.4.6 Performance Analytics Agent Module

This module tracks, analyzes, and visualizes campaign performance to measure ROI and improve future strategies.

1. Tracks campaign KPIs such as impressions, clicks, reservations, and conversions.
2. Applies sentiment analysis to detect customer satisfaction or negative feedback.
3. Generates dashboards and reports to visualize ROI and campaign effectiveness.

1.4.7 Admin Control Monitoring Module

This module provides administrators with full control over campaigns and ensures proper system monitoring.

1. Provides oversight of all running campaigns.
2. Allows pausing, resuming, or stopping campaigns if needed.
3. Maintains logs and alerts for errors or anomalies.

1.4.8 Authentication Security Module

This module ensures secure user registration, authentication, and controlled access to the system.

1. Manages user registration and login.
2. Provides role-based access (e.g., business user vs. system admin).
3. Ensures data security for stored customer and campaign information.

1.4.9 Database Management Module

This module manages structured and unstructured data related to businesses, campaigns, and customer feedback.

1. Stores structured data such as campaign details, performance metrics, and user profiles.
2. Stores unstructured data such as reviews, social media insights, and customer feedback.
3. Ensures consistency and reliability of stored information.

1.4.10 Integration API Module

This module connects the web app with external APIs and ensures smooth communication with AI/ML models.

1. Handles connectivity with external services (Meta Ads API, Google Ads API, WhatsApp Business API).

2. Manages interaction between the web app and AI/ML models.

3. Ensures smooth data flow between system modules and third-party platforms.

1.5 Work Division

Table 1.2: Work Division among Team Members

Team Member	Responsibilities
Ayyan	Architected the LangGraph state machine and orchestrated the multi-agent pipeline. Developed the **Strategy Planner** agent using GPT-4o and implemented the Real-time Agent Monitoring via Server-Sent Events (SSE).
Ali	Designed the React frontend dashboard and integrated the agentic workflow with the UI. Engineered the complex prompt chains for the **Trend Scout** and **Content Writer** agents to ensure context retention.
Hamza	Configured the FastAPI backend and SQLAlchemy database models. Developed the **Visual Designer** agent using the DALL-E 3 API and implemented the **Campaign Publisher** agent for automated data persistence.

Chapter 2

Project Requirements

This section outlines the necessary requirements for the successful completion of the project. Project requirements can be divided into two main categories: functional requirements, which describe the system's core operations, and non-functional requirements, which specify performance, security, and usability standards.

2.1 Use Cases

The TakhleeqX system primarily revolves around users setting up campaigns and the system autonomously executing them. The following are the key use cases:

- **UC-1: User Registration & Onboarding** - A new user signs up, verifies their account via JWT, and provides business details (cuisine, target demographic, tone) to tailor the AI generation.

- **UC-2: Campaign Launch** - The user initiates a new campaign from the dashboard. The system triggers the LangGraph state machine, invoking the five-agent pipeline sequentially.

- **UC-3: Real-Time Monitoring** - While the campaign is generating, the user watches real-time updates streamed via Server-Sent Events (SSE) indicating which agent is currently active.

- **UC-4: Review Generated Content** - Once the Campaign Publisher finishes, the user views the generated hashtags, captions, strategic pillars, and DALL-E 3 images in a simulated social feed.

2.2 Functional Requirements

The functional requirements describe the core behaviors of the TakhleeqX platform.

2.2.1 User Interface & Onboarding Module

1. **FR-UI-1:** The system shall allow users to securely register and log in using JWT-based authentication.
2. **FR-UI-2:** The system shall provide an onboarding form to capture business parameters, saving them to the SQLite database via SQLAlchemy.
3. **FR-UI-3:** The system shall display a simulated social feed rendering text, hashtags, and images returned by the backend.

2.2.2 Agentic Orchestration Module

1. **FR-AO-1:** The system shall implement a directed graph using LangGraph to sequence the 5 distinct AI agents.
2. **FR-AO-2 (Trend Scout):** The system shall employ a Trend Scout agent to autonomously fetch and analyze current marketing trends relevant to the user's domain.
3. **FR-AO-3 (Strategy Planner):** The system shall utilize a Strategy Planner agent, powered by GPT-4o, to generate a cohesive campaign strategy based on the scouted trends.
4. **FR-AO-4 (Content Writer):** The system shall feature a Content Writer agent that dynamically generates targeted captions, hashtags, and textual content tailored to the campaign strategy.
5. **FR-AO-5 (Visual Designer):** The system shall integrate a Visual Designer agent leveraging DALL-E 3 to autonomously create high-quality hero images corresponding to the generated text.
6. **FR-AO-6 (Campaign Publisher):** The system shall use a Campaign Publisher agent to handle data persistence and mock the final publishing phase into the simulated feed.
7. **FR-AO-7:** The system shall pass the state (JSON object) reliably between these agent nodes in the graph without data loss.

8. **FR-AO-8:** The system shall emit Server-Sent Events (SSE) to the frontend whenever any agent begins or completes its specific task.

2.3 Non-Functional Requirements

2.3.1 Reliability

The system relies on external APIs (OpenAI). It shall gracefully handle API rate limits or downtime by alerting the user rather than crashing. The SQLite database should maintain data integrity using SQLAlchemy's transactional models.

2.3.2 Usability

The user interface must be intuitive, built with Tailwind CSS v4, requiring zero technical knowledge of AI prompts from the end user. The system abstracts all complex prompt engineering behind simple form inputs.

2.3.3 Performance

PER-1: The SSE connection must establish within 1 second of campaign launch, providing immediate feedback to the user to prevent perceived system hangs during long LLM generations. **PER-2:** The frontend dashboard shall render within 2 seconds under normal network conditions.

2.3.4 Security

SEC-1: User passwords must be securely hashed using bcrypt before storage. **SEC-2:** API endpoints (except login/register) must be protected and require a valid JWT token in the Authorization header.

Chapter 3

System Overview

Give a general description of the functionality, context, and design of your project. Provide any background information if necessary.

3.1 Architectural Design

The TakhleeqX platform follows a modern, decoupled client-server architecture, further layered with a micro-agentic backend. The frontend acts as the presentation layer built with React, communicating with the FastAPI backend via RESTful endpoints and Server-Sent Events (SSE) for real-time updates. The backend orchestrates the business logic using LangGraph, which manages the stateful transitions between five distinct AI agents.

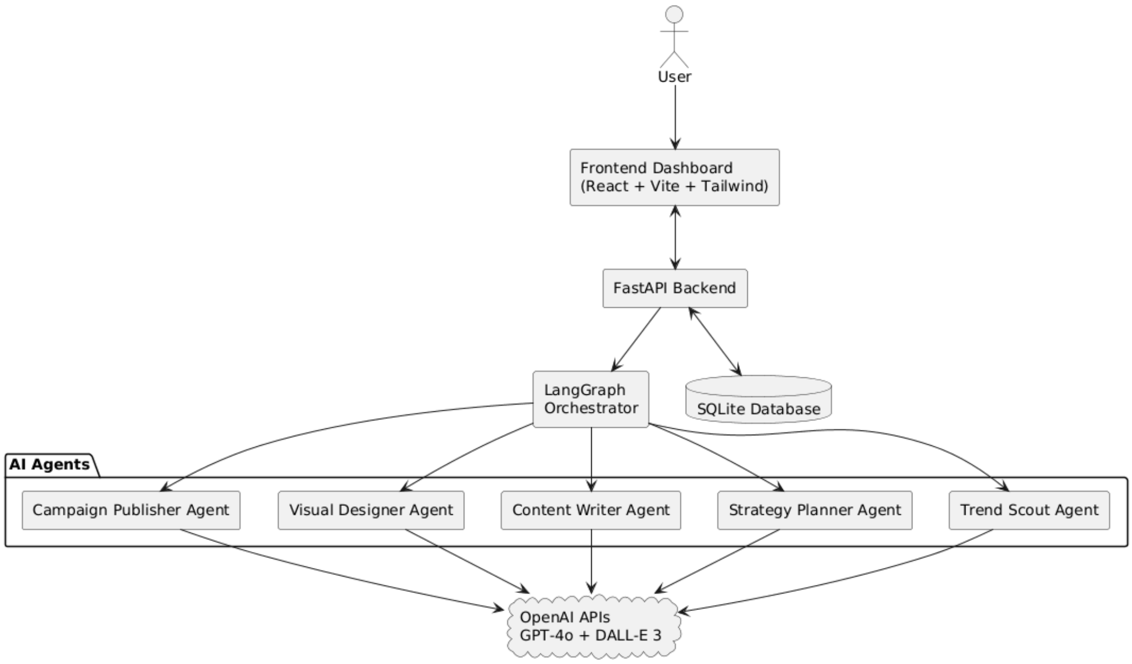


Figure 3.1: Box and Line Diagram of TakhleeqX

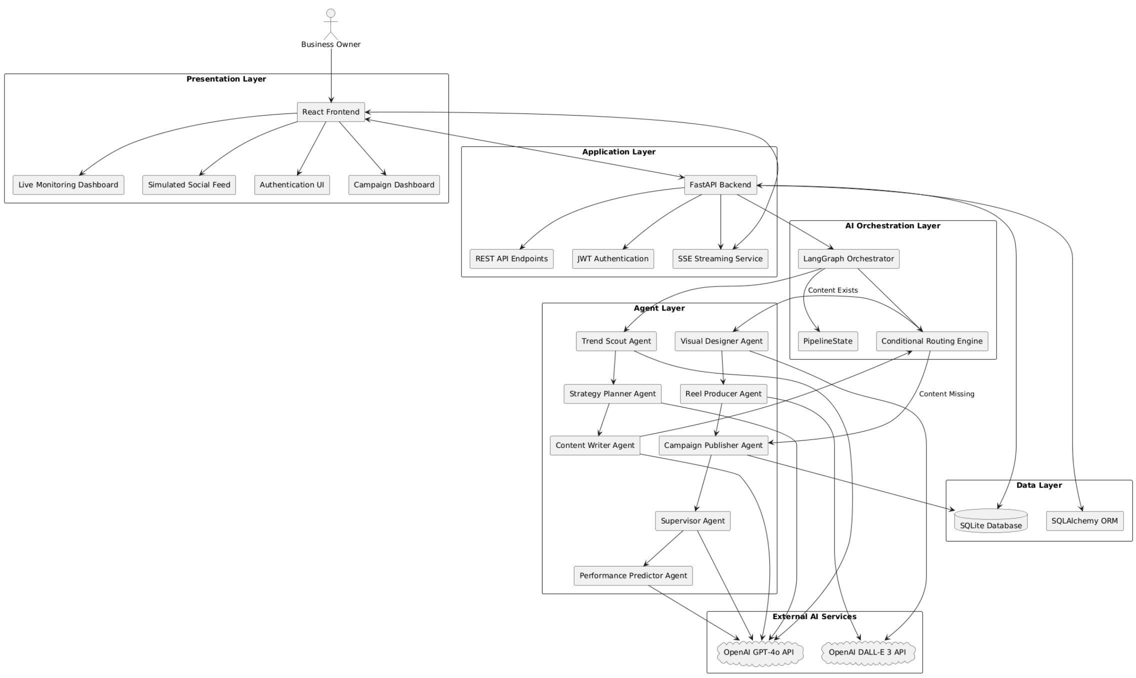


Figure 3.2: Detailed System Architecture showing LangGraph interactions

3.2 Design Models

The system is modeled using an Object-Oriented approach, capturing the dynamic behavior of the AI agents and the static structure of the database entities.

3.2.1 Activity Diagram

The activity diagram illustrates the sequential flow of control from the moment a user launches a campaign, through the LangGraph agent pipeline, to the final publishing step.

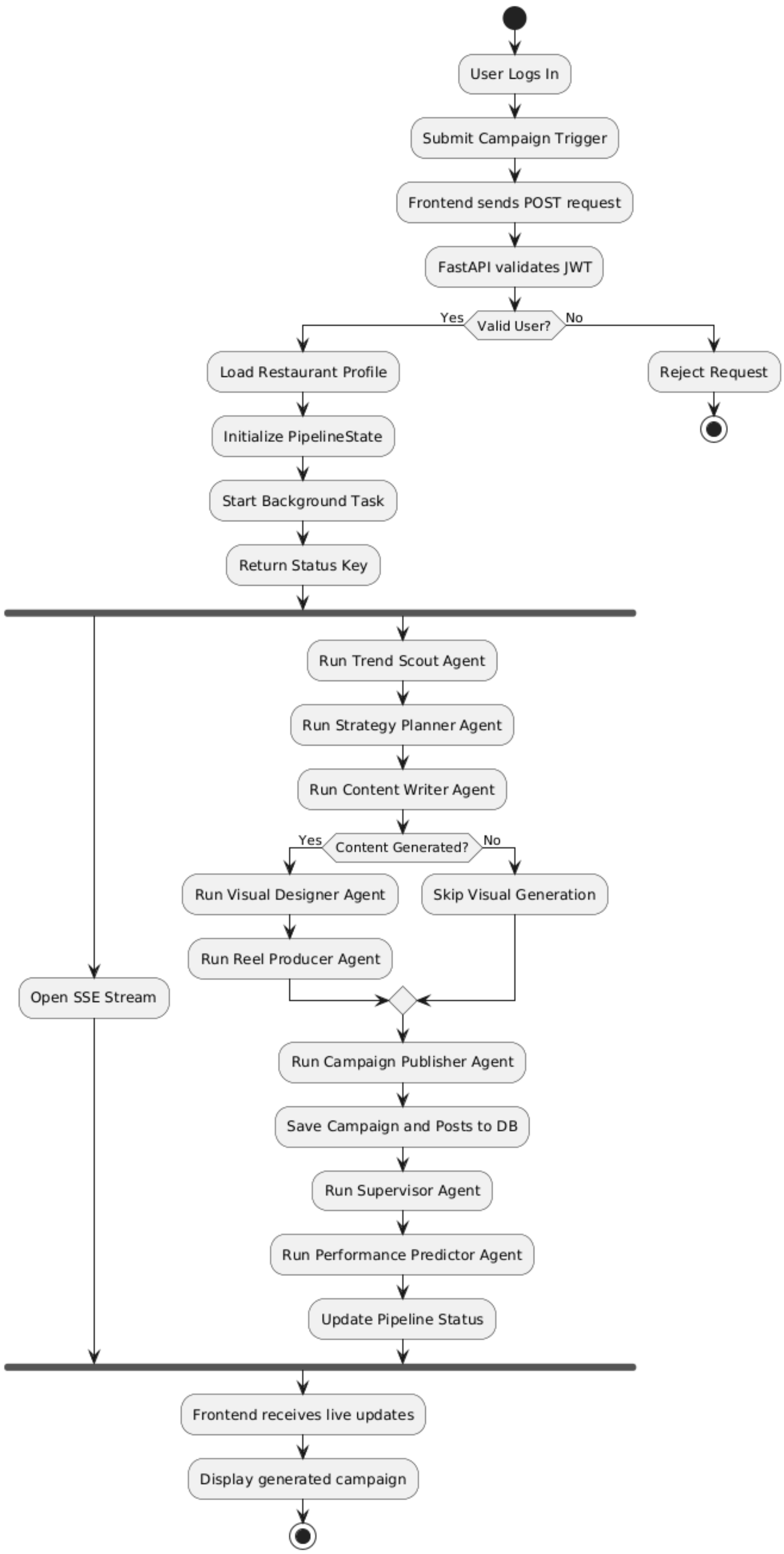
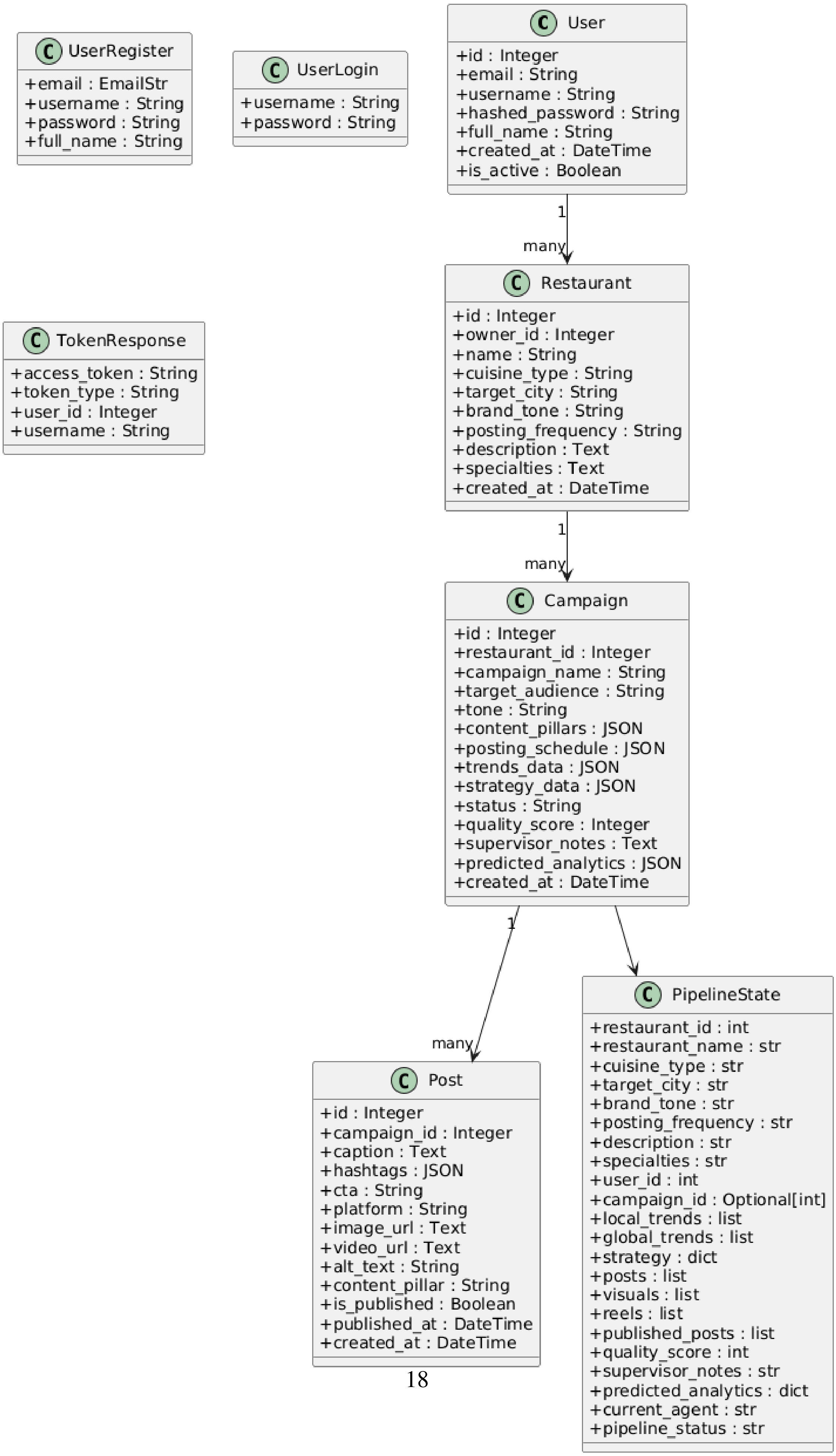


Figure 3.3: Activity Diagram for Campaign Generation

3.2.2 Class Diagram

The class diagram shows the static structure of the system, including the `User`, `Campaign`, and `Post` models, as well as the Agent classes.


UserRegister
+email : EmailStr
+username : String
+password : String
+full_name : String
UserLogin
+username : String
+password : String
User
+id : Integer
+email : String
+username : String
+hashed_password : String
+full_name : String
+created_at : DateTime
+is_active : Boolean
1
many
Restaurant
+id : Integer
+owner_id : Integer
+name : String
+cuisine_type : String
+target_city : String
+brand_tone : String
+posting_frequency : String
+description : Text
+specialties : Text
+created_at : DateTime
TokenResponse
+access_token : String
+token_type : String
+user_id : Integer
+username : String
1
many
Campaign
+id : Integer
+restaurant_id : Integer
+campaign_name : String
+target_audience : String
+tone : String
+content_pillars : JSON
+posting_schedule : JSON
+trends_data : JSON
+strategy_data : JSON
+status : String
+quality_score : Integer
+supervisor_notes : Text
+predicted_analytics : JSON
+created_at : DateTime
1
many
Post
+id : Integer
+campaign_id : Integer
+caption : Text
+hashtags : JSON
+cta : String
+platform : String
+image_url : Text
+video_url : Text
+alt_text : String
+content_pillar : String
+is_published : Boolean
+published_at : DateTime
+created_at : DateTime
PipelineState
+restaurant_id : int
+restaurant_name : str
+cuisine_type : str
+target_city : str
+brand_tone : str
+posting_frequency : str
+description : str
+specialties : str
+user_id : int
+campaign_id : Optional[int]
+local_trends : list
+global_trends : list
+strategy : dict
+posts : list
+visuals : list
+reels : list
+published_posts : list
+quality_score : int
+supervisor_notes : str
+predicted_analytics : dict
+current_agent : str
+pipeline_status : str

3.2.3 Sequence Diagram

The sequence diagram demonstrates the message passing between the React frontend, the FastAPI backend, and the external OpenAI APIs during the execution of the LangGraph state machine.

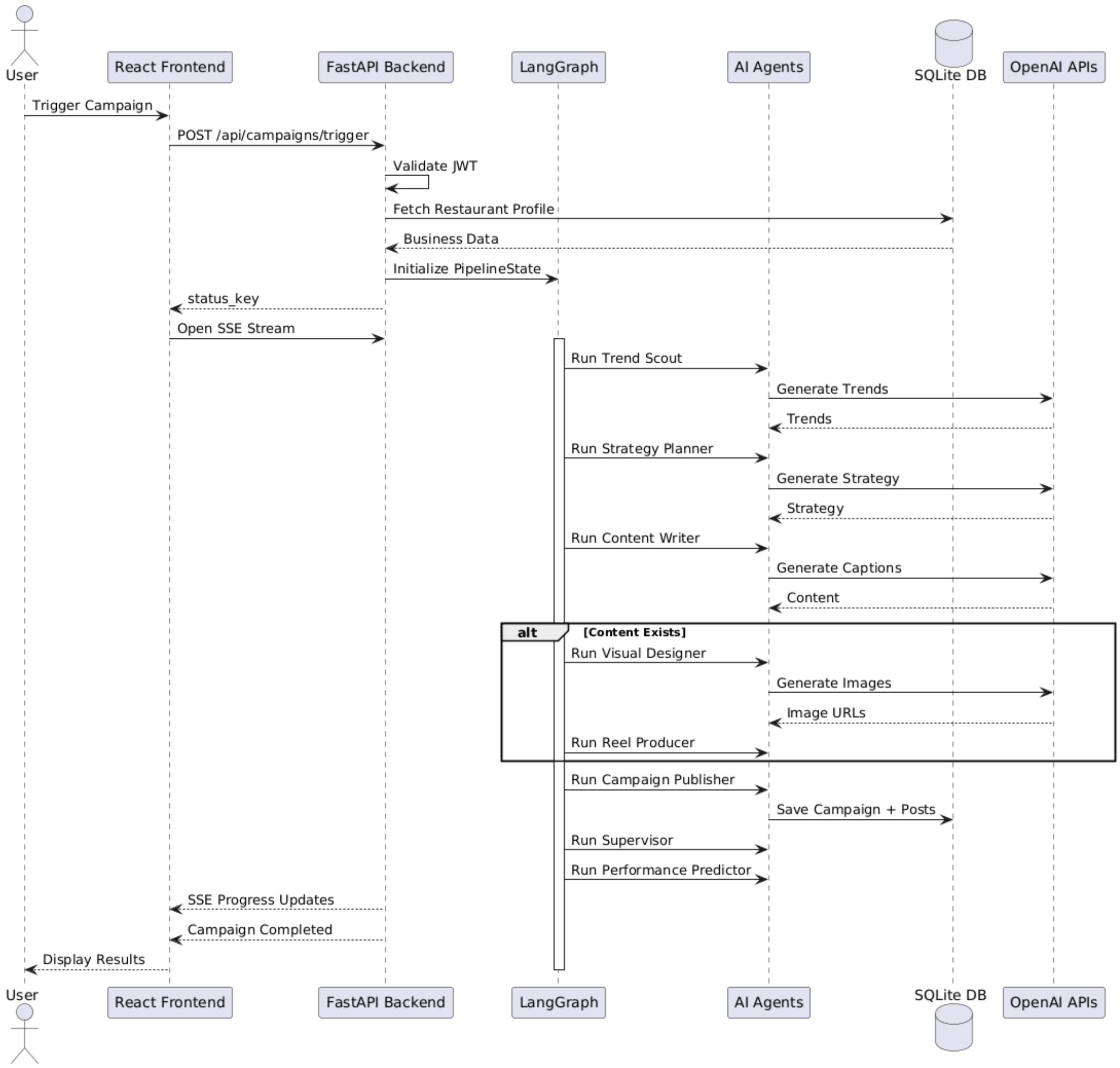


Figure 3.5: Sequence Diagram for Agent Orchestration

3.2.4 Data Flow Diagrams (DFD)

While TakhleeqX relies heavily on object-oriented and state machine models, Data Flow Diagrams are included to illustrate the high-level movement of information between the user, the core processing system, and external AI services.

Level 0 (Context Diagram): Shows the TakhleeqX platform as a single process interacting with the SME User and the OpenAI API.

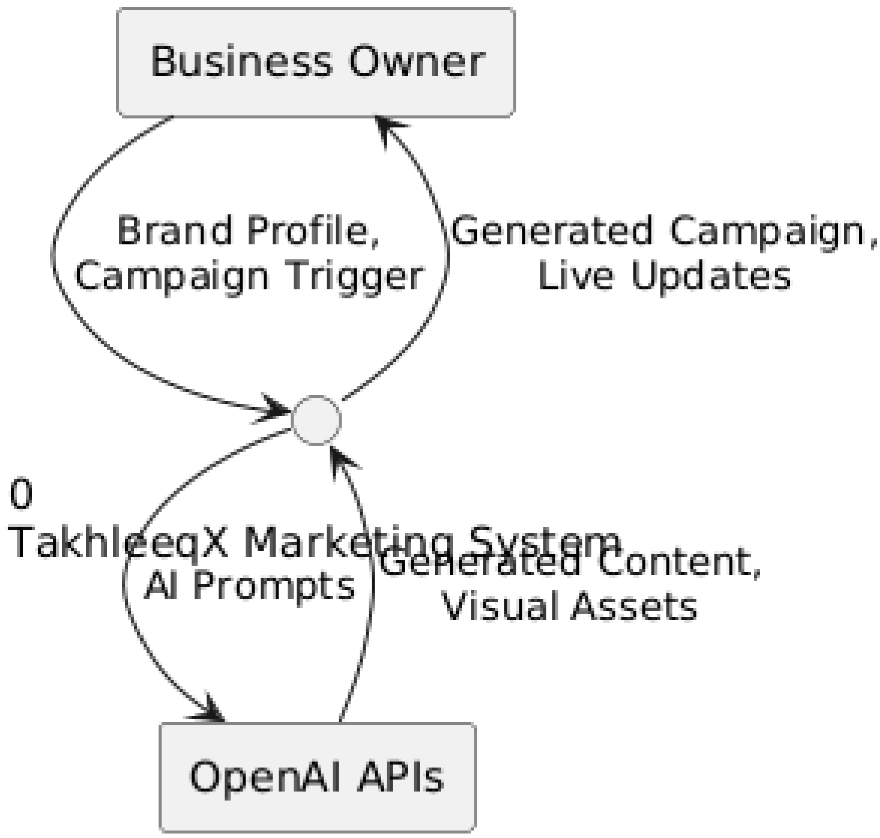


Figure 3.6: DFD Level 0 (Context Diagram)

Level 1 DFD: Breaks down the central platform into sub-processes including user authentication, business profile management, AI pipeline orchestration, and database persistence.

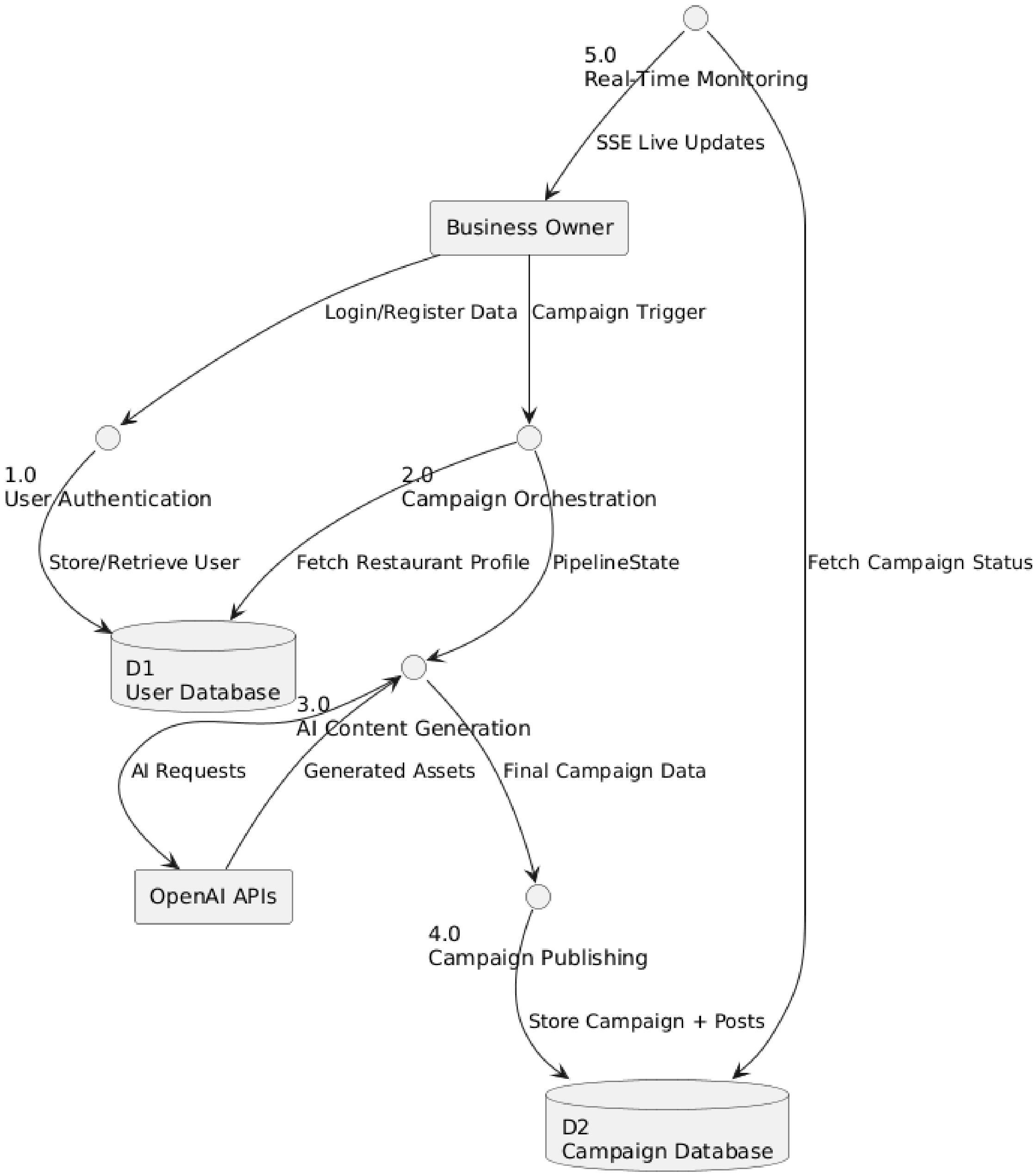
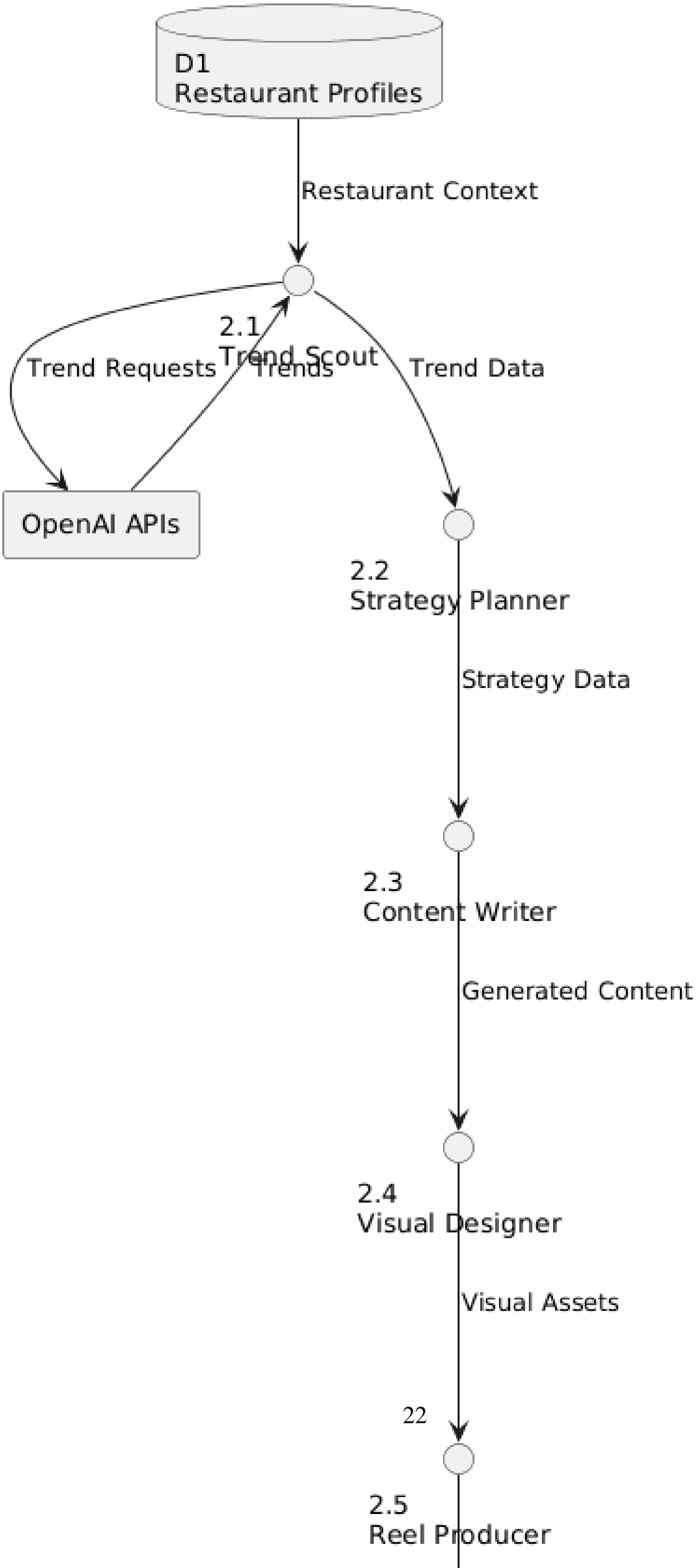


Figure 3.7: DFD Level 1

Level 2 DFD (AI Orchestration): Expands the "AI Pipeline Orchestration" process to detail the sequential data flow between the internal agents (Trend Scout, Strategy Planner, Content Writer, Visual Designer, and Campaign Publisher) and the shared state.


D1
Restaurant Profiles
Restaurant Context
2.1
Trend Scout
Trend Requests
Trends
Trend Data
OpenAI APIs
2.2
Strategy Planner
Strategy Data
2.3
Content Writer
Generated Content
2.4
Visual Designer
Visual Assets
2.5
Reel Producer

3.3 Data Design

TakhleeqX utilizes an SQLite database managed via SQLAlchemy (an Object-Relational Mapper). The database ensures persistent storage of user credentials, business profiles, and generated campaign assets. The primary entities are:

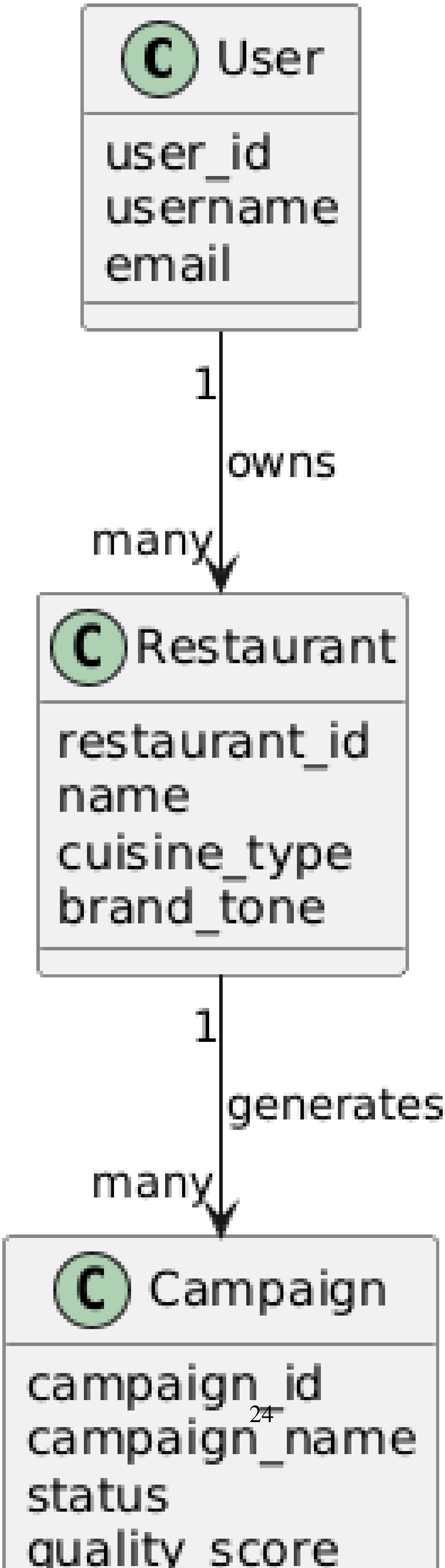
- **Users**: Stores hashed passwords and basic profile information.

- **Business Profiles**: Stores the parameters (cuisine, tone) that guide the AI agents.

- **Campaigns & Posts**: Stores the output of the agents, including image URLs and generated captions.

3.4 Domain Model

The domain model provides a conceptual view of the marketing automation domain, highlighting the core entities that TakhleeqX interacts with.


C User
user_id
username
email
1
owns
many
C Restaurant
restaurant_id
name
cuisine_type
brand_tone
1
generates
many
C Campaign
campaign_id
campaign_name
status
quality_score

The domain model illustrates how a User owns multiple Campaigns, and how each Campaign consists of several generated Posts (which include an image generated by DALL-E and text generated by GPT-4o). The system bridges these concepts by automating the creation process that would normally require manual intervention by a human marketing team.

Chapter 4

Implementation and Testing

Give a general description of the functionality, context, and design of your project. Provide any background information if necessary.

4.1 Algorithm Design

The core logic of TakhleeqX relies on a state machine algorithm orchestrated by LangGraph. This ensures that the context (business profile, trend data) is properly mutated and passed from one agent to the next without loss.

Pseudocode for Agent Orchestration Pipeline:

```
1. Initialize State Graph with schema (business_name, tone, trend, content, image)
2. Add Nodes: [TrendScout, StrategyPlanner, ContentWriter, VisualDesigner, Publishe
3. Define Edges:
    - START -> TrendScout
    - TrendScout -> StrategyPlanner
    - StrategyPlanner -> ContentWriter
    - ContentWriter -> VisualDesigner
    - VisualDesigner -> Publisher
    - Publisher -> END
4. Compile Graph
5. Function ExecuteCampaign(user_input):
6.    state = user_input
7.    for node in Graph.stream(state):
8.        emit(node.name, "Processing...")
9.        state = node.output
10.       emit(node.name, "Completed.")
```

```
11.   return state
```

4.2 External APIs/SDKs

TakhleeqX relies heavily on state-of-the-art generative AI APIs to power its autonomous agents.

Table 4.1: Third-Party APIs Used

API and version	Description	Purpose of usage	API endpoint/function used
OpenAI API (GPT-4o)	Large Language Model	Text generation, Strategy formulation, Trend analysis	`ChatOpenAI.invoke()`
OpenAI API (DALL-E 3)	Image Generation Model	Creating visual assets for campaigns	`client.images.generate()`
LangChain Core	LLM Orchestration SDK	Providing abstractions for prompts and outputs	`PromptTemplate, StrOutputParser`

4.3 Testing Details

Rigorous testing was conducted to ensure the reliability and functional accuracy of TakhleeqX, focusing especially on the unpredictable nature of generative AI outputs.

4.3.1 Unit Testing

Unit tests were implemented for the FastAPI backend using `pytest`. Individual endpoints, such as the user registration (`POST /auth/register`) and campaign creation routes, were tested in isolation using an in-memory SQLite database to mock the production environment. Furthermore, each AI agent function (e.g., `call_trend_scout`) was tested with mock LLM responses to ensure the state graph correctly routes the output without requiring expensive external API calls during routine CI/CD pipelines.

4.3.2 Integration Testing

Integration testing involved executing the full LangGraph pipeline from start to finish, validating that the JSON state emitted by the Visual Designer successfully parsed into the database schema expected by the Campaign Publisher. We also verified the Server-Sent Events (SSE) connection by simulating a client listening for status updates and asserting that the backend correctly streams the agent transitions.

Chapter 5

Conclusions and Future Work

5.1 Conclusion

TakhleeqX successfully demonstrates the viability and immense potential of multi-agent AI systems in the realm of digital marketing. By orchestrating five specialized agents (Trend Scout, Strategy Planner, Content Writer, Visual Designer, and Campaign Publisher) using LangGraph, the platform abstracts away the complexities of campaign generation. The integration of modern web technologies (React, FastAPI, SSE) with advanced generative models (GPT-4o, DALL-E 3) resulted in a seamless, autonomous pipeline. This project effectively solves the problem of high overhead and fragmentation in marketing workflows, offering SMEs a unified, user-friendly tool to maintain a dynamic and engaging online presence with minimal human intervention.

5.2 Future Work

While the current iteration of TakhleeqX is fully functional in a simulated environment, several avenues remain for future enhancement:

- **Live Social Media Integration:** The most critical next step is integrating official APIs from platforms like Meta (Facebook/Instagram), LinkedIn, and Twitter to automatically publish the generated content directly to live accounts.
- **Human-in-the-loop (HITL) Workflows:** Adding a review layer where users can edit or approve AI-generated content before the final publishing step to ensure maximum brand safety.
- **Advanced Analytics:** Implementing a closed-loop feedback system where real-world post performance metrics (likes, shares, conversions) are fed back into the

Strategy Planner agent to continuously optimize future campaigns.

- **Video Content Generation:** Expanding the Visual Designer's capabilities to include short-form video generation APIs (e.g., Sora or HeyGen) for platforms like TikTok and Instagram Reels.

Bibliography

[1] A Kolyshkin and S Nazarovs. Stability of slowly diverging flows in shallow water. *Mathematical Modeling and Analysis*, 2007.